

Luminal regulation of normal and neoplastic human EC cell serotonin release is mediated by bile salts, amines, tastants, and olfactants

Mark Kidd,¹ Irvin M. Modlin,¹ Bjorn I. Gustafsson,¹ Ignat Drozdov,¹ Oyvind Hauso,¹ and Roswitha Pfragner²

¹Gastrointestinal Pathobiology Research Group, Yale University School of Medicine, New Haven, Connecticut; and ²Institute of Pathophysiology and Immunology, Centre for Molecular Medicine, Medical University of Graz, Graz, Austria

Submitted 1 February 2008; accepted in final form 9 June 2008

Kidd M, Modlin IM, Gustafsson BI, Drozdov I, Hauso O, Pfragner R. Luminal regulation of normal and neoplastic human EC cell serotonin release is mediated by bile salts, amines, tastants, and olfactants. *Am J Physiol Gastrointest Liver Physiol* 295: 260–272, 2008. First published June 12, 2008; doi:10.1152/ajpgi.00056.2008.—Mechanisms by which gut luminal content regulates secretion and motility are ill understood. We evaluated whether neuroendocrine enterochromaffin (EC) cells act as luminal sensors for a wide variety of nutrients and defined the secretory mechanisms of this process. Pure (98–99%) FACS-sorted human EC cells and neoplastic EC cells (KRJ-I) were studied. RT-PCR identified transcripts for T2R1 (bitter), OR1G1 (class II olfactory) and trace amine (TAR1) G protein-coupled receptors (GPCRs) and transporters for glutamine (SNAT1/2), glucose (GLUT1/3/SGLT1), and bile salts (ABST). Glutamine and sodium deoxycholate stimulated 5-HT release (EC_{50} = 0.002–0.2 μ M; 2-fold release) but were 10–100 times more potent in neoplastic EC cells, which also secreted 6–13 times more 5-HT. Tastants (caffeine, tyramine, octopamine) and olfactants (thymol and eugenol) also stimulated normal and neoplastic EC cell 5-HT secretion (EC_{50} = 1.2 nM to 2.1 μ M and 0.05 nM to 0.1 μ M release, respectively); 2-deoxyglucose and the artificial sweetener sucralose also stimulated (EC_{50} = 9.2 and 0.38 nM). 5-HT release was associated with ERK phosphorylation (1.5-fold, P < 0.02) and could be inhibited by a somatostatin analog (IC_{50} = 1 pM). Eleven secretory associated genes including the vesicle docking inhibitor STXBP3 were upregulated in response to glutamine and bile salt stimulation in neoplastic EC cells. Targeting STXBP3 expression by use of antisense knockdown significantly (P < 0.05) reduced 5-HT secretion. In conclusion, EC cells express GPCRs and transporters for luminal tastants, olfactants, glutamine, glucose, and bile salts. Activation includes a panel of secretory genes, ERK phosphorylation, and 5-HT secretion. Luminal EC cell regulation is likely to be as important as G cell regulation in gastric acid secretion; development of agents to target EC cell function is therefore a critical therapeutic goal.

gastrointestinal; neuroendocrine; olfactant; secretion; tastant; enterochromaffin cells

THE ENTEROCHROMAFFIN (EC) cell is the predominant neuroendocrine cell of the gastrointestinal (GI) tract and plays a key role in the regulation of secretion, motility, and visceral pain. The majority (95%) of serotonin (5-HT) in the body is in intestinal EC cells (24), which synthesize, store, and release this amine (1, 5, 11, 30, 59, 60). Electron microscopic studies of EC cells demonstrate a polarized cell with apical microvilli that protrude into the intestinal lumen. 5-HT is located in secretory granules at the basolateral pole and via paracrine, neurocrine, and hormonal release activates intestinal entero-

cytes, mucus cells, neurons, and smooth muscle receptors to regulate secretion, motility, and visceral pain perception (23, 58). Serotonin thus appears to be of fundamental importance to intestinal function, akin to the pivotal role of gastric histamine, mucosal levels which correlate almost directly with gastric acid secretion (21).

Physiological 5-HT secretion is associated with peristalsis, gastric motility, and postprandial pancreatic secretion (13, 22, 28, 67, 88), whereas abnormalities in 5-HT release and availability (reuptake and catabolism) are associated with altered secretion and motility resulting in diarrhea, constipation, and pain, as seen in the carcinoid syndrome (56, 81), irritable bowel syndrome (17, 22), and inflammatory bowel disease (63). Central and enteric nervous system signaling represents a substantial regulatory element of 5-HT secretion (67), whereas digestion-related luminal stimuli (e.g., nutrients, tastants, olfactants, or bile salts) are thought to represent important local EC cell activators (23, 61, 64, 79, 88).

The function of the EC cell is essentially as a sensor of the gut luminal milieu (12, 26), with ingested agents or their breakdown products as effectors of 5-HT release that is then responsible for triggering secretory and motility reflexes (20, 88). Excessive release of 5-HT from EC cell carcinoids characteristically results in excessive fluid and electrolyte secretion and exaggerated peristalsis culminating in severe diarrhea, borborygmi, and abdominal pain (56, 57). Similarly, meal ingestion is often associated with exacerbation of both irritable bowel syndrome (33) and carcinoid symptoms (56) (paroxysmal diarrhea), suggesting that activation of EC cells by luminal agents is involved in such events. Somatostatin receptors (sst2/5) are expressed on EC cells, and targeting subtypes 2 and 5 is known to inhibit 5-HT secretion (4, 32). Whether somatostatin also inhibits luminally mediated 5-HT secretion is not known.

Although it is apparent how sampling the luminal contents of the stomach is critical to the ability of the G cell to regulate gastric secretion and thereby define gastric function, similar information is not available to understand how key small intestinal digestive events are modulated by neuroendocrine sensory cells sampling the lumen. Since the intestinal EC cell is the predominant neuroendocrine cell in the gut and its secretory products are key regulators of a wide variety of critical digestive functions (13, 22, 28, 67, 88), we sought to define which luminal agents regulate 5-HT secretion and identify the mechanistic basis for this process. Since animal and

Address for reprint requests and other correspondence: I. Modlin, Yale Univ. School of Medicine, TMP202, 333 Cedar St., New Haven, CT 06520 (e-mail: imodlin@optonline.net).

The costs of publication of this article were defrayed in part by the payment of page charges. The article must therefore be hereby marked “advertisement” in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

mucosal models are limited in terms of delineating the site specificity of lumenally mediated events, we developed a strategy to study normal human EC cell secretion. To characterize the regulatory events in neoplastic EC cells, we compared and contrasted our observations with a human neoplastic EC cell line. We propose that normal EC cells sense a variety of luminal signals and in response secrete 5-HT, which has ubiquitous functions in regulating the digestive process; similarly that neoplastic EC cells exhibit similar propensities but of a different sensitivity, thus accounting for the notorious paroxysmal response of carcinoid patients to certain ingested luminal agents.

Our ability to undertake such investigation has been facilitated by the recent development of pure rodent and human preparations of EC cells (44, 55). Prior to this, physiological information regarding the regulation of EC cell secretion in both health and disease reflected studies of a variety of intact animal preparations or in cell lines that are not EC cell derived (9, 42, 48, 65, 66). In the present study, we isolated and studied the secretory responses of EC cells from surgically resected human small intestinal mucosa and compared this to the responses in the neoplastic EC cell line KRJ-I (62) to test the hypothesis that neoplastic EC cells respond with different secretory profiles to normal EC cells.

Thereafter, we investigated potential mechanisms by which carcinoid tumors (neoplastic EC cells) could oversecrete 5-HT and "hyperreact" (paroxysmal response) to specific ingested agents (37, 49, 83, 87). To do this, we evaluated whether ERK signaling or calcium flux regulated 5-HT release and then determined whether the secretory and ERK-signaling effects could be reversed by using a somatostatin analog that targeted sst2/5 in neoplastic EC cells. Finally, we measured the secretory transcriptome in response to amino acids and bile salts and evaluated the effect of transcript knockdown on 5-HT secretion in neuroendocrine EC cells. These studies enabled us to mechanistically define luminal regulation of normal and neoplastic EC cell secretion, identify the regulatory role of dietary agents (e.g., tyramine; Ref. 8), and determine how 5-HT secretion could be pathologically perturbed in EC cell neoplasia.

MATERIALS AND METHODS

Immunocytochemistry. Formalin (4%)-fixed normal EC cells or neoplastic cells were rehydrated and treated with 3% hydrogen peroxide for 10 min to block endogenous peroxidase activity and incubated with primary antiserum for 2 h at room temperature (43). Polyclonal goat anti-SNAT1/2, ABST (Santa Cruz), or rabbit anti-TR1 (identifies TAS1R1/TAS1R2 and TAS1R3 receptors)/TR2 (identifies TAS2R1) (Alpha Diagnostic International) diluted (1:200) in PBS containing 0.25% Triton X-100 (Calbiochem) and 0.25% BSA (Sigma) was used to stain neuroendocrine cells. Cy5-tyramide (NEN Life Science Products)-labeled antibodies (Santa Cruz) were used to identify transporter and receptor immunoreactivity (43). Nuclei were stained with 4',6-diamidino-2-phenylindole (20 mg/ml).

Cell isolation. Normal EC cells were isolated from the ilea (macroscopically normal adjacent ilea from hemicolecotomies for colon cancer, $n = 6$) following mucosal stripping, enzymatic digestion, and a combination of Nycodenz gradient fractionation, acridine orange uptake, and FACS sorting as described (55). In previous studies, preparations of ~98–100% pure EC cells were obtained (55). Approximately 1×10^6 cells were obtained per mucosal sample, a quantity sufficient for short-term culture, secretion, and signaling studies (55).

Measurement of transcripts for glutamine and bile salt transporters and receptors for tastants and olfactants. Total RNA was extracted from normal small intestinal mucosa ($n = 5$), 5×10^5 isolated EC cells ($n = 5$) or 1×10^6 neoplastic cells ($n = 4$) using TRIzol (Invitrogen) extraction and cleaned up with the Qiagen RNeasy kit in conjunction with the DNeasy Tissue kit (Qiagen) to ensure no contaminating genomic DNA (55). RNA was converted to cDNA (High Capacity cDNA archive kit; Applied Biosystems). Real-time PCR was performed (ABI 7900 sequence detection system) in triplicate using Assays-on-Demand primers as described (55). Primers included SLC2A1/2/3 (GLUT1–3), SLC5A1 (SGLT1), SLC38A1/2, SLC10A1/2, TAR1, TAS1R1, R2, R3, TAS2R1, mGluR4, and O1R1. Gene expression levels were normalized by using geNorm (82) and expression of the novel house-keeping genes, *ALG9*, *TFCP2*, and *ZNF410* (46).

EC cell culture. FACS-sorted cells were divided into aliquots of $\sim 5 \times 10^4$ cells/100 μ l culture media (Ham's F12 medium, GIBCO) supplemented with FBS (10%) and antibiotics (100 U penicillin/ml + 100 μ g streptomycin/ml, Sigma-Aldrich), seeded into 96-well collagen I-coated plates (Becton Dickinson), and maintained in a humidified atmosphere (55).

Neoplastic EC cell (KRJ-I) cultures. Neoplastic EC cells were cultured in Ham's F12 medium (GIBCO) supplemented with 10% FBS and antibiotics (100 U penicillin/ml + 100 μ g streptomycin/ml). Cultures were maintained in a humidified atmosphere at 37°C in 5% CO₂. For experiments, cells (log phase growth, passage number 50–75) were plated at a density of 10^5 cells/well without antibiotics.

Secretion experiments. After FACS sorting and 2 h short-term culture, normal or neoplastic EC cells were stimulated with each agent (10^{-12} to 10^{-6} M). Isoproterenol (1 μ M), a β_1 -adrenergic stimulatory G protein-coupled receptor (GPCR) ligand and cAMP signaling ligand (43, 55), and forskolin (10 μ M), which is a cAMP activator, were used as positive controls. Cells were stimulated with physiological concentrations of glucose (2-deoxyglucose, sucralose), glutamine, the trace amines tyramine and octopamine, the bile salt sodium deoxycholate (SDC), the bitter tastant caffeine, and the olfactants eugenol and thymol (10^{-12} to 10^{-6} M) for 60 min as described (44, 48, 55). For somatostatin inhibition experiments, cells were preincubated with BIM23244 (sst2/5 analog: 10^{-12} to 10^{-9} M) for 15 min prior to 60-min stimulation with EC₅₀ concentrations of agents. These experiments were repeated in triplicate for each concentration. Media were collected and frozen at -80°C until ELISA analysis. To confirm that release represented a secretory process rather than cell damage or lysis, lactate dehydrogenase (LDH) release into media was measured (commercially available LDH assay, CytoTox-ONE Homogenous Membrane Integrity Assay, Promega) (55).

5-HT ELISA assay. 5-HT release was measured by a commercially available ELISA (BA 10-0900; Rocky Mountain Diagnostics). This assay is based on the competitive binding technique in which 5-HT present in a sample competes with a fixed amount of horseradish peroxidase-labeled 5-HT for sites on a mouse monoclonal antibody. The range for this assay is 15–2,500 pg/ml. We have previously successfully measured 5-HT release into media using this assay following the manufacturer's instructions (44, 55).

ERK phosphorylation and signaling studies. Neoplastic EC cells were either stimulated with the EC₅₀ concentration of each candidate agent for 5 min. Isoproterenol (1 μ M) and forskolin (10 μ M) were used as controls. For inhibitory studies, cells were preincubated with the sst2/5 analog BIM23244 for 15 min prior to 5 min of stimulation with each candidate agent. ERK signal activity was determined by SuperArray CASE ELISA kits (ERK; FE-002), which measure phosphorylation at T202/Y204 and have previously been used to measure ERK signaling in gastric ECL cells (45) and KRJ-I cells (47). Stimulated cells were fixed (4% formaldehyde) and stained with either phosphorylated or nonphosphorylated antibodies [60 min, room temperature (RT)]. After washing and secondary antibody application (60 min, RT), cells were incubated with color developer (10 min, RT) and

plates read at 450 nm. Protein was then be assayed in each well (reading at 595 nm) and used to normalize ERK signals.

Calcium flux measurements. Changes in intracellular calcium levels were measured by one-color flow cytometry as described (29). Neoplastic EC cells were resuspended in assay buffer (PBS) without Ca^{2+} or Mg^{2+} and treated with 2 μM of the acetoxymethyl ester of fluo-3 in 0.02% Pluronic F-127 and 2.5 μM probenecid for 20 min at RT, washed in assay buffer without $\text{Ca}^{2+}/\text{Mg}^{2+}$ containing probenecid, and resuspended in the same buffer at 5×10^5 cells. Immediately prior to experiments, PBS containing Ca^{2+} (36 mM)- Mg^{2+} (20 mM) was added, and after 5 min cells were analyzed on a FACScalibur flow cytometer (Becton Dickinson) at RT. Changes in fluo-3 fluorescence were measured in the FL1 (FITC) channel. Baseline fluorescence was determined for 2–3 min, after which samples were stimulated with the agents. Rightward shifts in FITC are considered reflective of Ca^{2+} influx.

Transcriptome analysis. Neoplastic EC cells were stimulated with glutamine (0.2 μM) or SDC (20 nM) for 60 min. RNA was extracted as described above from these cells or unstimulated cells (controls) and submitted to the Keck Affymetrix Resource (<http://info.med.yale.edu/wmkeck/affymetrix/>) at Yale University. Human EXON arrays (HuEX-ST Array) were used to measure global alterations in transcripts. Briefly, labeled cDNA was performed on 1 μg of total RNA following removal of ribosomal RNA (rRNA) utilizing the RiboMinus Human/Mouse Transcriptome Isolation Kit (Invitrogen). Double-stranded cDNA was synthesized with random hexamers tagged with a T7 promoter sequence; this was then used as a template for the subsequent amplification in the presence of T7 RNA polymerase, producing multiple antisense cRNA copies. In the second cycle of cDNA synthesis, random hexamers were used to reverse transcribe the cRNA to produce single stranded DNA in the sense orientation. To reproducibly fragment the single-stranded DNA, dUTP was incorporated in the DNA at this stage. The single stranded cDNA thus obtained was then fragmented by using the combination of uracil DNA glycosylase and apurinic/apyrimidinic endonuclease 1. The fragmented single-stranded DNA was subsequently labeled with biotin by use of Affymetrix proprietary DNA labeling reagent. Fragmented cDNA was hybridized to the arrays for 16 h at 45°C. After hybridization, arrays were washed via the Affymetrix fluidics station 450 and stained with streptavidin-phycoerythrin (10 $\mu\text{g}/\text{ml}$, Molecular Probes), and washed arrays were scanned on an Affymetrix GeneChip scanner 3000. Scanned output files were visually inspected for hybridization artifacts and then analyzed. Genes differentiating unstimulated (control) cells from SDC- and glutamine-stimulated cells were detected by use of Partek (59a). A gene-level analysis was performed

on natural log-transformed data, and geometric fold change ≥ 2.0 was considered to be significant. To reduce error associated with filtering, normalized gene expression of ≤ 5.0 was excluded and false discovery rate of 1% was used. A functional group analysis was used to identify categories of commonly altered genes utilizing the Gene Ontology terms (www.geneontology.org). Categories with < 5 genes or > 500 were removed from this analysis because they are either too specific or too general to be useful.

Transcript knockdown (siRNA) studies. We next evaluated the effects of 48-h knockdown of STXBP3 on neoplastic EC cell secretion, using a small interfering (si)RNA protocol (70). Briefly, neoplastic EC cells were plated in 96-well plates (2.5×10^4 cells/well) overnight and then transfected with RNA-lipid complexes (100 nM RNA, 0.4 $\mu\text{l}/\text{well}$ in DharmaFECT 2 or transfecting media with scrambled siRNA) or mock transfected (transfecting media alone) as described (70). Forty-eight hours after transfection, cells were stimulated for 60 min with isoproterenol (1 μM) or media alone (control-basal 5-HT secretion) and media was removed to measure secreted 5-HT and LDH levels (as an indicator of cell viability). RNA was isolated from the cells by use of TRIzol (as above) and the level of target knockdown was assessed by real-time PCR. ΔC_T values for STXBP3 were compared between the different groups. Transfection did not alter cell viability.

Data analysis. Results are expressed as means $\pm$ SD. All statistical analyses were performed via Prism 4 (GraphPad Software, San Diego, CA). Dose-response curves were calculated for each candidate secretory agent and the EC_{50} or IC_{50} was determined. Results were compared between control and stimulated cells by the Mann-Whitney test. A $P < 0.05$ was considered significant.

RESULTS

Demonstration and confirmation of transporters and taste- and odorant-responsive G protein-coupled receptor mRNA in normal and neoplastic EC cells. Real-time PCR of normal ($n = 5$) and neoplastic ($n = 4$) EC cells was undertaken to identify GPCRs and candidate luminal amino acid, glucose, and bile transporters that could be sensor molecules for these cells. Normal EC cells were characterized by transcripts for glutamine (SNAT1/2), glucose (GLUT1/3, SGLT1), tyrosine derivatives (TAR1), and the intestinal bile salt (ABST) transporters (Table 1). Expression of TAS1R1 and 1R2 were absent but TAS1R3 transcripts were identified, as was mGluR4. TAS1R3 and mGluR4 form heterodimers that recognize

Table 1. Transcript expression of candidate luminal transporters and G protein-coupled receptors in normal small intestinal mucosa, in isolated normal EC cells, and in KRJ-I cells

	Glutamine		Tyrosine	Bile Salts		Glucose			
	SLC 38A1	SLC 38A2	TAR1	SLC 10A1	SLC 10A2	SLC 2A1	SLC 2A2	SLC 2A3	SLC 5A1
<i>Amino acid, bile salt, and glucose transporters and trace amine G protein-coupled receptors</i>									
Normal mucosa	+	+	+	—	+	+	—	+	+
EC cells	+	+	+	—	+	+	—	+	+
KRJ-I	+	+	+	—	+	+	—	+	+
	Sweet			Umami		Bitter		Olfactants	
	TAS1R1	TAS1R2	TAS1R3	mGluR4		TAS2R1		O1R1	
<i>Taste and olfactory G protein-coupled receptors</i>									
Normal mucosa	—	—	+	+		+		+	
EC cells	—	—	+	+		+		+	
KRJ-I	—	—	+	—		+		+	

+, Present; —, absent; EC, enterochromaffin.

“umami” (51); the absence of 1R1 and 1R2 suggests that EC cells potentially will not recognize a “sweet” taste. In addition, the bitter (TAS2R1) taste receptor and olfactant receptor O1R1 that identifies odorants of different chemical classes (72) was identified. The expression profile in neoplastic EC cells was identical except for the absence of the umami receptor mGluR4.

Immunocytochemical confirmation of transporter and taste/odorant receptors in normal and neoplastic EC cells. To confirm that transcript expression resulted in protein expression that was specific to the EC cell, we evaluated candidate transporter and taste receptor expression in FBS-fixed normal EC cells and KRJ-I cells. Using this approach (43), we identified that both SNAT1 and SNAT2 as well as ABST were expressed in normal EC cells (Fig. 1, *top*). An antibody that identified all the subtypes of the TR1 receptor stained the cells, demonstrating protein expression of TAS1R3. TR2 labeling was also identified. Neoplastic EC cells also expressed protein for all three transporters and taste receptors (Fig. 1, *bottom*), results that confirm the real-time PCR experiments.

Stimulation of 5-HT secretion in normal and neoplastic EC cells by glutamine and the bile salt SDC. Prior to examining whether 5-HT can be stimulated or inhibited by candidate luminal agents, we compared basal 5-HT secretion between normal and neoplastic EC cells to evaluate whether there were any intrinsic alterations in neoplastic EC cell secretion. We demonstrated that basal 5-HT secretion from cultured normal human EC cells was $\sim 5 \times 10^{-8}$ M/h (55), levels adequate for effective in vivo activation of downstream secretory and neural pathways (2, 7, 10, 14, 16, 74, 75). In the KRJ-I cell line, basal 5-HT release ranged between $3\text{--}7 \times 10^{-7}$ M/h representing 6–13 $\times$ more 5-HT release than normal EC cells per equivalent cell concentration (5×10^4 cells). This indicates that the neoplastic EC cell releases substantially more 5-HT than normal EC cells and suggests that, in addition to increased EC cell mass, uncontrolled exocytosis may be one of the features of carcinoid neoplasia.

Glutamine stimulated both normal and neoplastic serotonin secretion with EC_{50} s of 4×10^{-8} and 4×10^{-9} M, respectively (Fig. 2A). The bile salt SDC stimulated secretion with

estimated EC_{50} s of 2×10^{-10} and 2×10^{-8} M, respectively (Fig. 2B). These results demonstrate that luminal agents, including an amino acid transported via SNATs (glutamine) (52) and bile salts, stimulate serotonin secretion. To exclude the possibility that the secretory effect of SDC occurred through cell damage, we measured LDH levels (60 min). SDC did not increase LDH (range alteration: -0.5% to $+1.3\%$ above baseline), indicating that secretion does not reflect a cytotoxic effect.

Stimulation of 5-HT secretion in normal and neoplastic EC cells. The secretory effect of trace amines, olfactants, and tastants was next examined. The trace amines tyramine and octopamine significantly stimulated KRJ-I serotonin secretion with EC_{50} s of 1×10^{-7} and 1×10^{-8} M ($\sim 3.2 \pm 0.3$ -fold, $P < 0.004$ vs. unstimulated cells) (Fig. 3, A and B). In normal EC cells, these amines stimulated secretion with EC_{50} 's of 2.1×10^{-7} and 2.1×10^{-6} M ($P < 0.05$ vs. unstimulated cells) (Fig. 3, A and B). The olfactants thymol and eugenol stimulated secretion with EC_{50} s of 6×10^{-8} and 1×10^{-6} M (neoplastic cells) and 2.9×10^{-7} and 1.4×10^{-7} M (normal cells), respectively (Fig. 3, C and D).

The prototypic bitter tastant caffeine induced a significant stimulation of 5-HT secretion in neoplastic cells (EC_{50} of 5.6×10^{-11} M vs. 1.2×10^{-8} M in normal EC cells, Fig. 4A). Receptors (TAS1R2) required to identify the prototypic sweet tastant glucose were not present on KRJ-I cells, but two glucose transporters, GLUT1 and GLUT3, and the Na^+ -glucose transporter SGLT1 were expressed (Table 1), suggesting EC cells may respond to sweet tastants or artificial sweeteners (53). 2-Deoxyglucose (Fig. 4B) stimulated normal 5-HT secretion with an EC_{50} of 9.2×10^{-9} M but inhibited neoplastic cell secretion ($IC_{50} = 2.5 \times 10^{-10}$ M) whereas sucralose (Fig. 4C) stimulated both normal and neoplastic 5-HT secretion with EC_{50} s of 3.8×10^{-10} and 2.4×10^{-9} M, respectively.

These results demonstrate that luminal agents including tastants and olfactants activate serotonin secretion but that the response to sweet tastants or artificial sweeteners is different in normal and neoplastic EC cells.

Demonstration of ERK signaling in response to candidate secretagogues. ERK signaling is a marker of tastant-mediated neuroendocrine secretion (38) and is associated with phosphor-

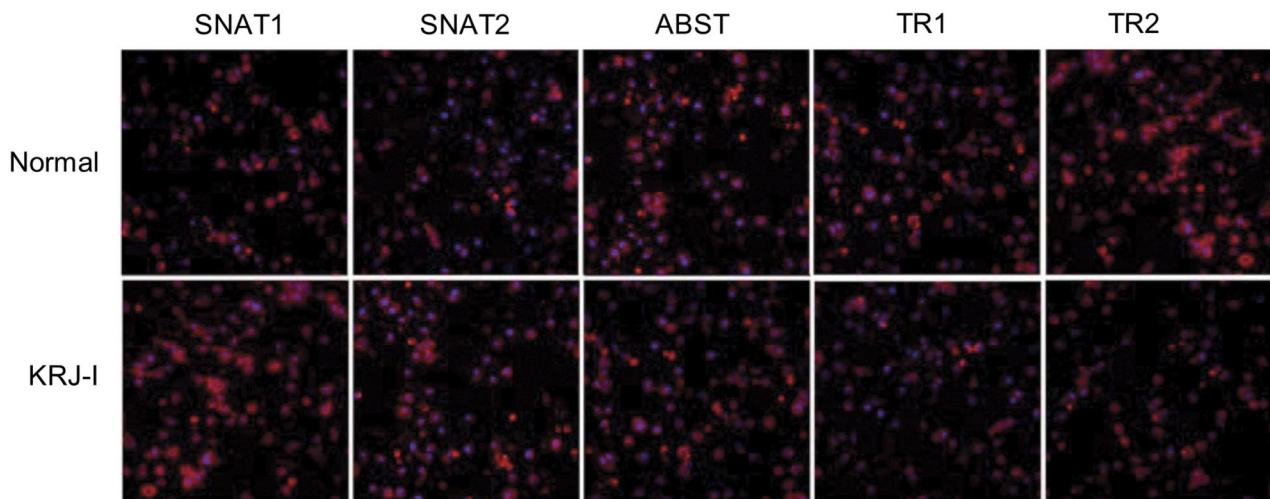
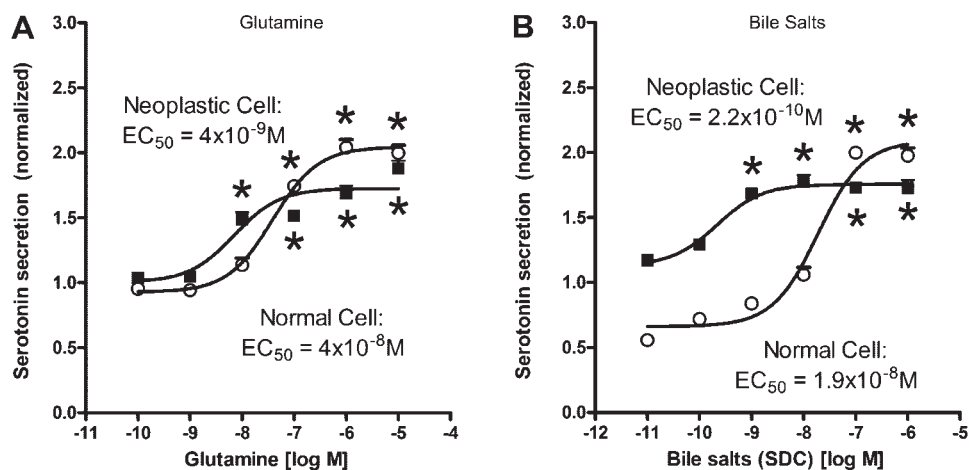


Fig. 1. Immunocytochemical identification of the glutamine transporters SNAT1/2, the intestinal bile salt transporter (ABST), and taste receptors (TR1/2) in normal and neoplastic enterochromaffin (EC) cells. 4',6-Diamidino-2-phenylindole (DAPI) was used for nuclei and Cy-5 for target proteins. Magnification $\times 400$.

Fig. 2. Glutamine and bile salt dose-response curves for serotonin secretion. The EC_{50} for glutamine on normal EC cells is 4×10^{-8} M and on neoplastic cells is 4×10^{-9} M (A). EC_{50} s for the bile salt, sodium deoxycholate, is 1.9×10^{-8} and 2.2×10^{-10} M, respectively (B). ■, Neoplastic EC cell; ○, normal EC cell. Values are means $\pm$ SD, $n = 3$. * $P < 0.05$ vs. unstimulated cells.

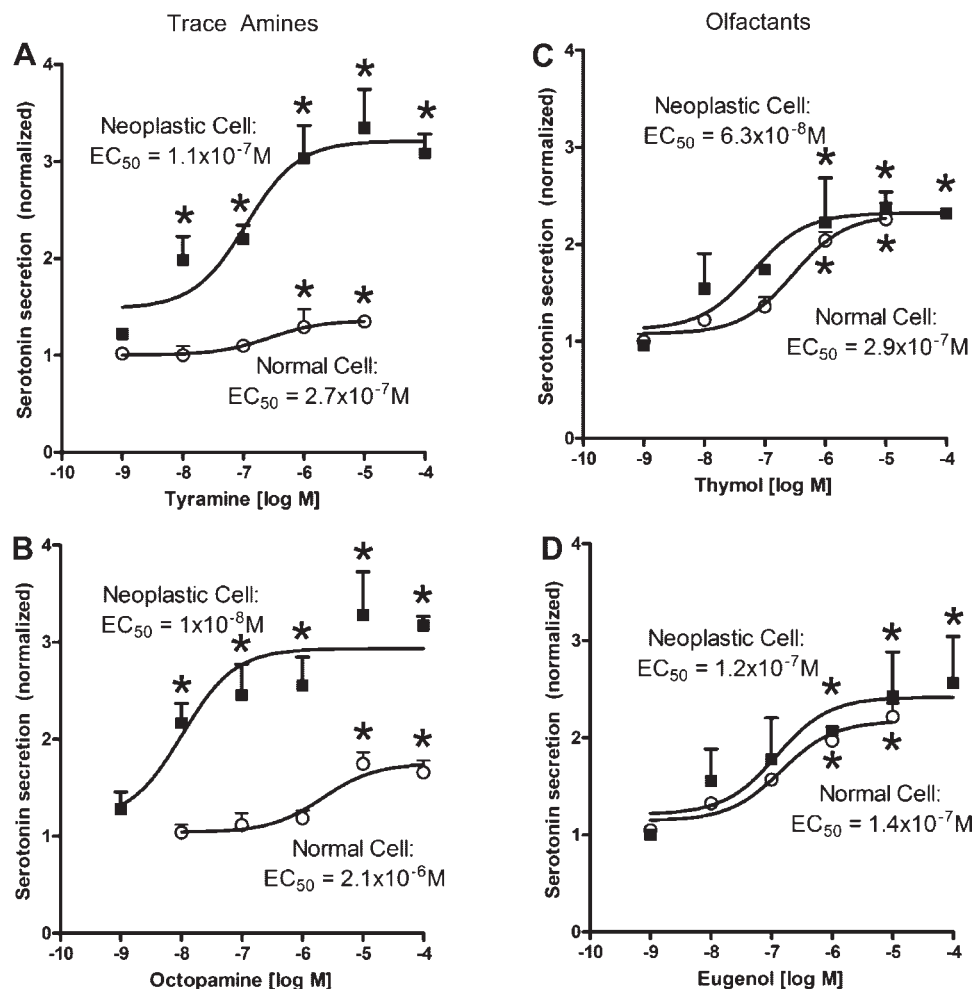


ylation, which can be identified by an ELISA approach (45). ERK phosphorylation in response to isoproterenol (1 μ M) and forskolin (10 μ M) (both stimulate neoplastic EC cell secretion through cAMP activation) (43) demonstrated a 15–18% increase ($P < 0.05$ vs. unstimulated cells) in phosphorylated ERK, signifying that cAMP stimulation activated signaling through the MAPK pathway (Fig. 5). Both tyramine and

octopamine significantly elevated pERK (12–30%, $P < 0.05$) as did glutamine (12%), SDC (16%), caffeine (19%), and sucralose (21%). Neither of the olfactants eugenol or thymol significantly altered ERK phosphorylation.

Demonstration of calcium signaling in response to candidate secretagogues. We next measured alterations in intracellular calcium fluxes using fluoro-3AM (29) to evaluate

Fig. 3. In normal and neoplastic EC cells, respectively, the EC_{50} for tyramine is 2.7×10^{-7} and 1×10^{-7} M (A) and for octopamine is 2.1×10^{-6} and 1×10^{-8} M (B). The EC_{50} for the olfactant thymol is 2.9×10^{-7} and 6×10^{-8} M (C) and for eugenol is 1.4×10^{-7} and 1.2×10^{-7} M (D). Values are means $\pm$ SD, $n = 3$. * $P < 0.05$ vs. unstimulated cells.



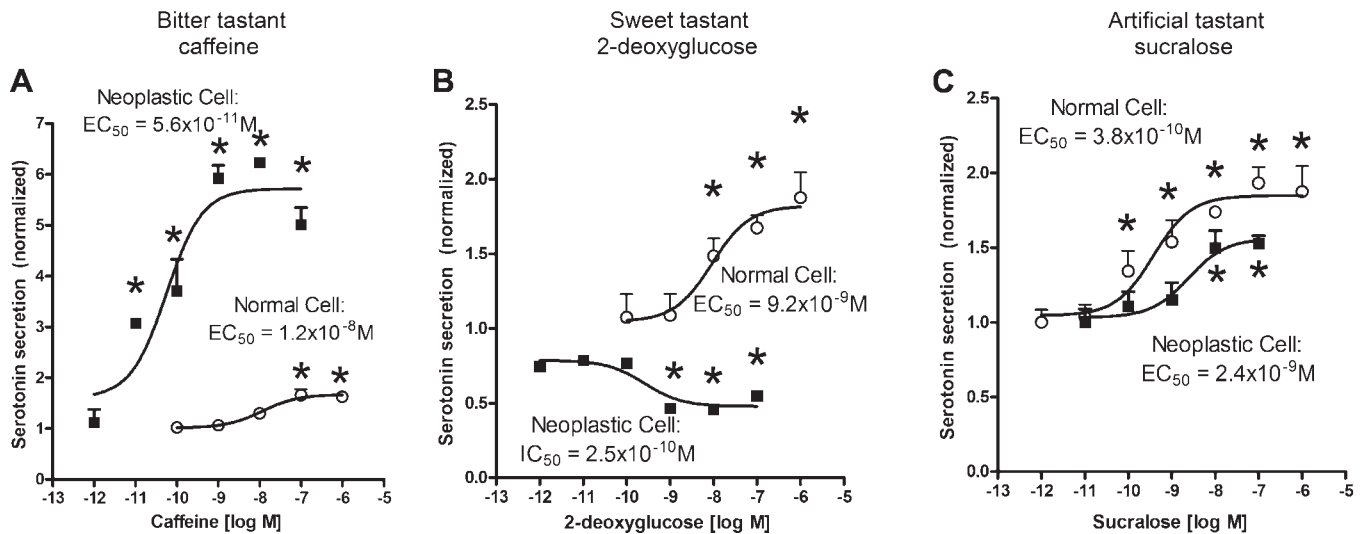


Fig. 4. In normal and neoplastic EC cells, respectively, the EC_{50} for the bitter tastant caffeine is 1.2×10^{-8} and 5×10^{-11} M (A). The EC_{50} for the sweet tastant 2-deoxyglucose was 9.2×10^{-9} M for normal EC cells and the IC_{50} was 2.5×10^{-10} M in neoplastic cells (B). The EC_{50} for the artificial sweet tastant, sucralose, was 3.8×10^{-10} and 2.4×10^{-9} M (C). Values are means $\pm$ SD, $n = 3$, $*P < 0.05$ vs. unstimulated cells.

whether calcium influx was involved in 5-HT secretion. Neither β_1 -adrenergic receptor activation (isoproterenol) nor forskolin was associated with significant calcium influx measured over 3 min in KRJ-I cells. In addition, tyramine, octopamine, glutamine, and SDC did not cause calcium influx. In contrast, both eugenol and thymol, which activates OR1G1, and caffeine, which is a bitter tastant, was associated with a significant (>5 -fold increase) in calcium influx in EC cells (Fig. 6).

Examination of inhibitory effects of somatostatin on candidate secretagogue 5-HT secretion and ERK phosphorylation. We next measured whether activation of sst2/5 inhibited luminally mediated serotonin release and ERK phosphorylation. Preincubation with BIM23244 significantly ($P < 0.05$) inhibited serotonin release by all four tested agents (glutamine, tyramine, caffeine, and thymol) with IC_{50} s in the picomolar range (2×10^{-13} to 3×10^{-12} M) (Fig. 7). Complete inhibition (not different from basal) was noted for all somatostatin analog concentrations >10 pM. These results demonstrate that a somatostatin analog can reverse amino acid-, tastant-, and olfactant-activated 5-HT secretion. Fifteen-minute

incubation with BIM23244 (10 pM) significantly (30%, $P < 0.05$) decreased ERK phosphorylation compared with controls (Fig. 8). Preincubating cells with this analog also significantly ($>50\%$, $P < 0.01$) reduced ERK phosphorylation in glutamine-, tyramine-, caffeine-, and thymol-stimulated cells, indicating that a component of the inhibitory effects of somatostatin is mediated through inhibition of ERK signaling.

Identification of candidate secretory KRJ-I genes upregulated by glutamine or SDC stimulation. Gene transcripts from neoplastic cells stimulated with glutamine (0.2 μ M) or SDC (20 nM) were measured on HuEx-1.0 Exon arrays and consistently overexpressed genes identified by using Partek Genomics Suite software. This identified 305 genes (of 17,890 genes) that were altered by both glutamine and SDC. Eighty-seven genes were upregulated and 218 genes were downregulated. By use of the Gene Ontology program, 11 of the 87 upregulated genes (13%) could be associated with a secretory phenotype (Table 2). These genes encoded for proteins associated with initiation of protein synthesis (EIF4A2), protein processing (N-glycosylation; ALG6/8), protein folding and transport across the ER (PDIA5, TRAM1, RAB1A), membrane target-

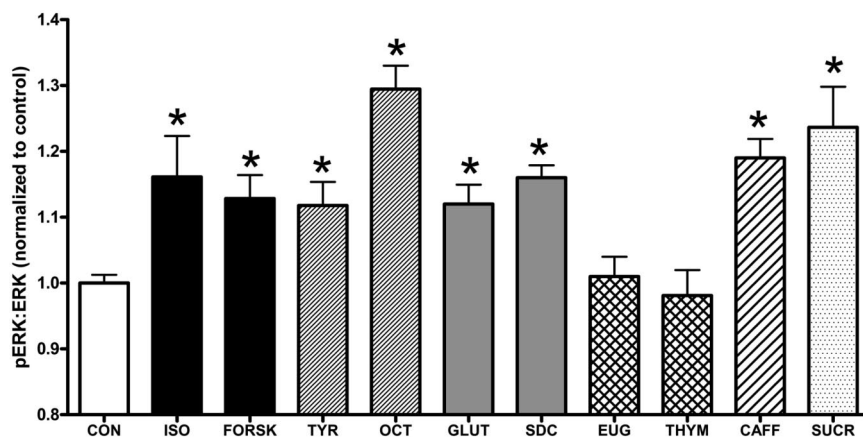
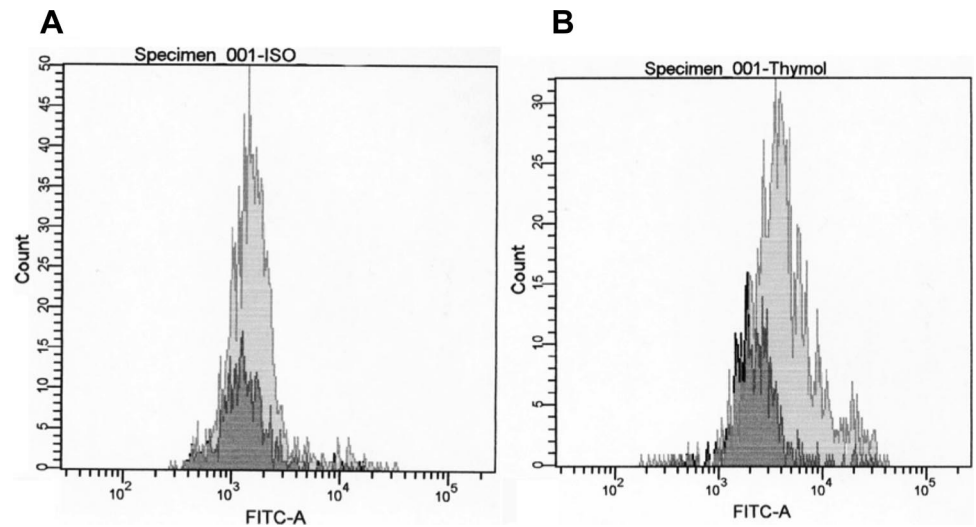


Fig. 5. Effect of candidate secretagogues on ERK phosphorylation in neoplastic EC cells. Isoproterenol (1 μ M; ISO) and forskolin (10 μ M; FORSK) significantly stimulated pERK. Tyramine (0.1 μ M; TYR), octopamine (10 nM; OCT), glutamine (4 nM; GLUT), sodium deoxycholate (0.2 nM; SDC), caffeine (60 pM; CAFF), and sucralose (2.4 nM; SUCR) all significantly stimulated ERK phosphorylation. Eugenol (1.5 μ M; EUG) and thymol (60 nM; THYM) had no effect. Means $\pm$ SE, $n = 4$, $*P < 0.05$ vs. control (CON).

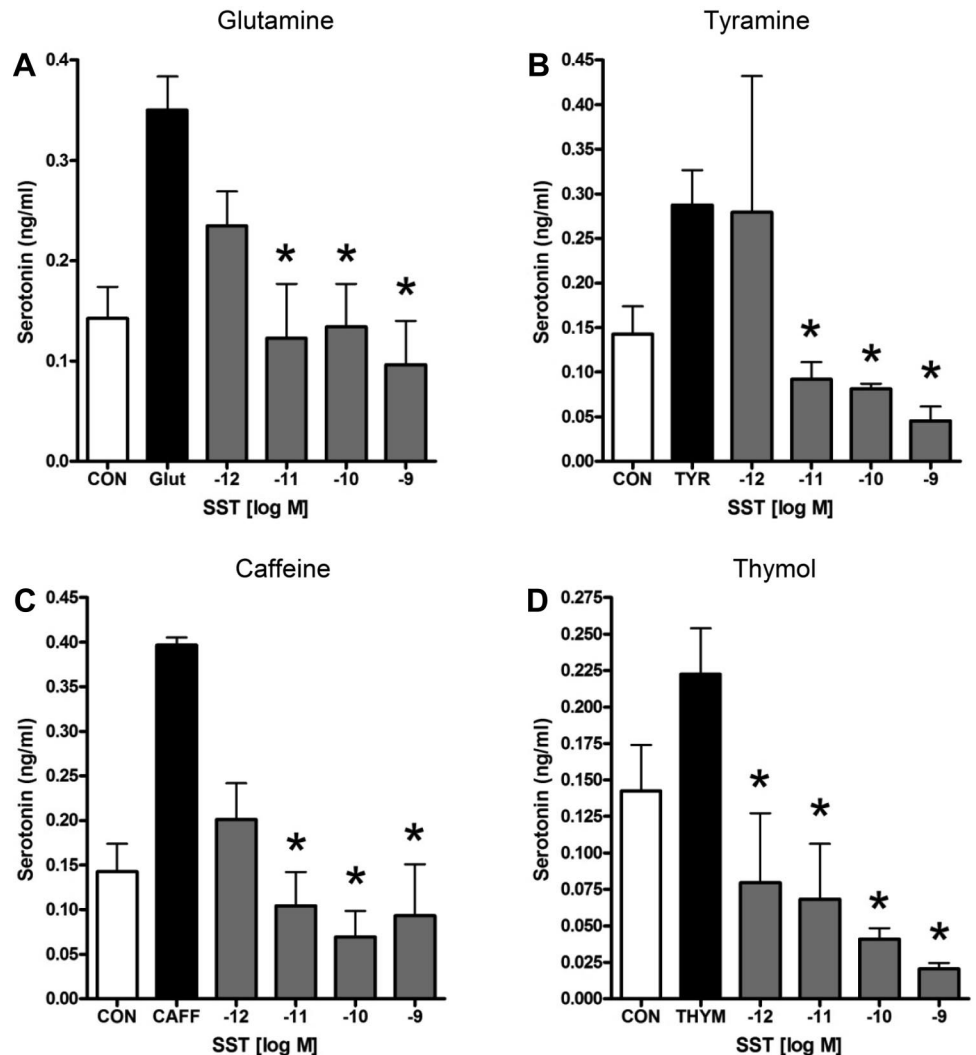
Fig. 6. Effect of isoproterenol (1 μ M; *A*) and thymol (60 nM; *B*) on 3-min intracellular Ca^{2+} flux in neoplastic EC cells. Thymol stimulated calcium influx (FITC shift: >5 -fold rightward), which was not observed following isoproterenol (FITC shift <1.5 -fold rightward). Similar (>5 -fold rightward shifts in FITC) was noted for eugenol (1.5 μ M) and caffeine (60 pM) but was not seen for forskolin (10 μ M), tyramine (0.1 μ M), octopamine (10 nM), glutamine (4 nM), and SDC (0.2 nM).



ing (SRP54), vesicle docking (STXBP3), and recycling (SFT2D1), functions integral for cell secretion. In addition, a gene (PAIP2) stabilizing aliphatic amino acid (GLUT) transporter mRNA was also upregulated.

Assessment of the effects of transcript knockdown on 5-HT secretion. To assess the mechanistic relevance of these genes to secretion, we evaluated the effects of knocking down STXBP3 using a siRNA approach on 5-HT release. Targeting

Fig. 7. Somatostatin (SST) analog (BIM23244) dose response curves for luminally mediated serotonin secretion in neoplastic cells. The IC_{50} for BIM23244 on glutamine (4 nM)-mediated secretion is 3×10^{-12} M (*A*), for tyramine (0.1 μ M)-mediated secretion 2×10^{-12} M (*B*), for caffeine (60 pM)-mediated secretion 5×10^{-13} M (*C*) and for thymol (60 nM)-mediated secretion 2×10^{-13} M (*D*). Values are means $\pm$ SD, $n = 3$, $*P < 0.05$ vs. control.



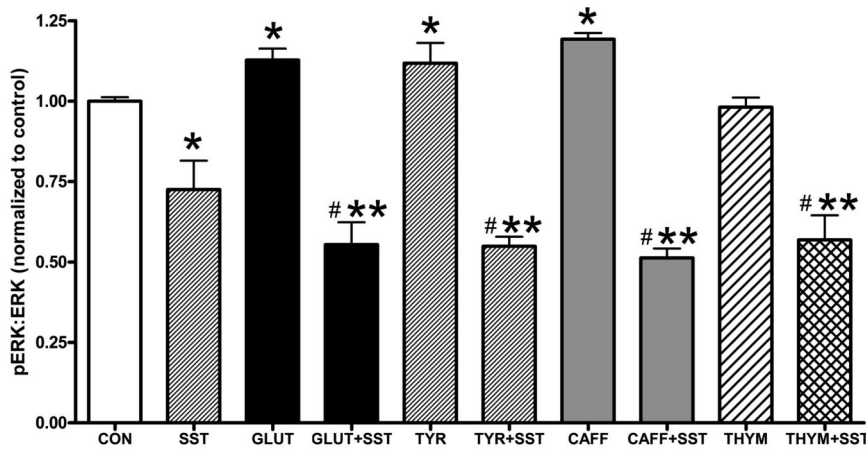


Fig. 8. Effect of somatostatin preincubation on candidate secretagogue-mediated ERK phosphorylation in neoplastic EC cells. Somatostatin (10 pM) inhibited ERK phosphorylation by 30% compared with control (unstimulated cells). Somatostatin significantly (>50%) inhibited glutamine (4 nM)-, tyramine (0.1 μ M)-, and caffeine (60 pM)-mediated ERK phosphorylation. Somatostatin also inhibited ERK phosphorylation in thymol (60 nM)-stimulated cells. Means $\pm$ SE, $n = 4$, * $P < 0.05$ vs. control (stimulation), # $P < 0.05$ vs. control (inhibition), ** $P < 0.01$ vs. stimulated cells.

STXBP3, an inhibitor of vesicle docking and exocytosis, increased both basal and β_1 -adrenergic-stimulated 5-HT secretion (Fig. 9).

DISCUSSION

A number of GI neuroendocrine cells, including GIP gene expression in rat intestine (31), CCK release from human intestinal and mouse STX1 cells (84), and GLP release from L-type cells (38), have been investigated in terms of functional responses to luminal effectors. However, the dominant neuroendocrine cell of the intestine, namely the 5-HT-secreting EC cell, which has a key role as a gut sensor (23, 61, 64, 79, 88), has not previously been well investigated in either its normal or neoplastic form (38, 78). In this investigation, we have examined the role of EC cells as responders to luminal or absorbed nutrients, tastants, and olfactants to evaluate both the functional and mechanistic role in the process of luminal sensing. In the present study, using normal human EC cells and the KRJ-I cell line, we demonstrate that 1) the EC cell expresses transporters for glutamine, bile salts, and glucose and G protein-coupled receptors for trace amines, tastants, and olfactants; 2) neoplastic cells secrete/release 6–13 times as much 5-HT as normal cells; 3) stimulation of normal and neoplastic cells with glutamine and SDC activates 5-HT release; 4) trace amines, caffeine, and olfactants activate 5-HT secretion from normal and neoplastic cells, and 2-deoxyglucose and sucralose stimulate 5-HT secretion; 5) secretion induced by glutamine and trace amines is associated with ERK phosphorylation; 6) calcium influx is related to tastant- and olfactant-mediated 5-HT secretion;

7) a somatostatin analog can inhibit luminally mediated secretion through ERK signaling inhibition; 8) 11 genes were upregulated by both glutamine and bile salts and constitute a component of the secretory phenotype of the neoplastic EC cell; and 9) knockdown of transcription of STXBP3 from this secretory transcriptome significantly reduced 5-HT secretion.

In previous studies, we have developed the protocol to isolate normal human EC cells (98–99%), characterized KRJ-I to be a neoplastic EC cell line, and demonstrated that EC cells synthesize and secrete 5-HT (55). β_1 -Adrenergic agents stimulate EC cell secretion via cAMP signaling and ERK activation whereas neural agents including acetylcholine and somatostatin inhibit these effects (43, 55). In the present study, we evaluated whether normal and neoplastic EC cells were sensitive to luminal or absorbed agents and examined the mechanisms of 5-HT secretion in these cells.

In normal and neoplastic EC cells, we identified transcript and protein expression for amino acids (glutamine and trace amines), bile salts, glucose and candidate taste (bitter/umami), and the class II (terrestrial type) olfactant receptor O1R1 that is also expressed in nasal epithelia and is a general receptor that identifies odorants of different chemical classes (72). This suggests that EC cells can “taste and smell” the luminal environment of the bowel and can respond to luminal agents (e.g., caffeine, trace amines, or bile salts) through expression of specific GPCRs and transporters.

Glutamine is considered a critical amino acid in terms of its ability to stimulate processes involved with gut mucosal healing (76). It is also a regulator of the GABA response and a

Table 2. Details of 11 selected genes upregulated in both glutamine and bile salt-stimulated EC cells

Transcript ID	Accession Number	Gene Assignment	Function	P Value	Fold Alterations
2656738	NM_001967	EIF4A2	Protein synthesis	<0.01	>2
2339682	NM_013339	ALG6	N-glycosylation	<0.001	>2
3383164	NM_024079	ALG8	N-glycosylation	<0.01	>2
3602723	NM_002902	RCN2	ER Ca ²⁺ binding	<0.01	>2
2639225	NM_006810	PDIA5	Protein folding and chaperone	<0.005	>2
3139882	NM_014294	TRAM1	Transport - ER	<0.0002	>2
2556667	NM_004161	RAB1A	Transport - ER	<0.005	>2
3532313	NM_003136	SRP54	Targeting to membrane	<0.01	>2
2350339	NM_007269	STXBP3	Vesicle docking Exocytosis	<0.01	>2
2984573	NM_145169	SFT2D1	Vesicle fusion	<0.001	>2
2831209	NM_016480	PAIP2	mRNA stability	<0.001	>2

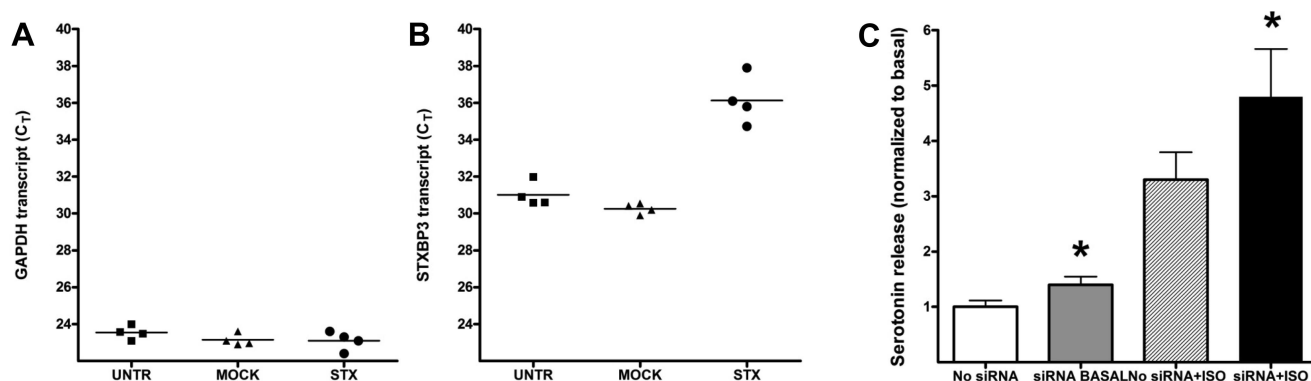


Fig. 9. Effect of knockdown of the STXBP3 (STX) mRNA in neoplastic EC cells. Addition of STXBP3 small interfering (si)RNA for 48 h to EC cells had no effect on GAPDH transcript levels (A) but significantly decreased STXBP3 transcripts (B). Untreated cells (UNTR) and mock-transfected cells (MOCK) showed no alterations in ΔC_T values for this gene. STXBP3 knockdown increased both basal and isoproterenol (1 μ M)-mediated serotonin release (C). Values are means $\pm$ SE, $n = 3$. * $P < 0.05$ vs. no siRNA.

well-known modulator of gut proliferation, differentiation, and immune function (86). Bile salt uptake in the ileum is critical for the enterohepatic circulation (27). Little, however, is known about the effects of either glutamine or bile salts on EC cell function. In the present study, we demonstrated that both agents stimulated 5-HT secretion and ERK signaling. The half-stimulatory concentrations for both agents were significantly lower in neoplastic EC cells compared with the normal EC cells.

The effects of glutamine have previously been examined on GLP-1 secretion from the stable immortalized murine GLUTag cell line (68). The estimated half-maximal stimulatory effect was 10^{-4} to 10^{-3} M, which is ~ 5 – 6 logs higher concentrations than for normal and neoplastic EC cells. This difference suggests that cells from the ileum are more sensitive to amino acids than colonic cells (GLUTag was isolated from a SV40-driven colon tumor expressing glucagon; Ref. 50), that the production of the cell line may have altered the secretory phenotype of these murine cells, or it may reflect the intrinsic differences between the regulatory roles of serotonin (neural activator; nanomolar concentrations; Ref. 22) and glucagon (cell activator; millimolar concentrations; Ref. 68) in GI physiology. Amino acid uptake is also associated with GLP-1 secretion from the human colon carcinoma NCI-H716 cell line following ERK phosphorylation (69). Although neoplastic EC cell 5-HT secretion occurs through the same pathway, the differences between these two models are highlighted by the large difference in stimulatory effects: nanomolar concentrations for KRJ-I and millimolar concentrations in NCI-H716 cells.

The bile salt SDC causes small intestinal serotonin secretion, a phenomenon that could be blocked by L-type calcium channel antagonists (61). In the present study, we identified that normal cells respond to bile salts with 5-HT release and that a transformed EC cell is more sensitive to these ligands than normal EC cells. The latter provides one possible explanation for bile salt-induced diarrhea (77), one feature of the GI symptoms associated with carcinoids (57). A comparison with the literature (61) demonstrates that normal EC cells *in vitro* are $\sim 1,000$ times more sensitive to SDC than cells *in situ*. This reflects either the effects of inhibitory cells on EC cell function *in situ* or technical difficulties with stimulating EC cells in this model. Bile acids, lithocholic acid, and deoxycholic acid stimulate GLP-1 release from STC-1 cells through a cAMP/ERK

pathway (41). The estimated half-maximal effect is in the micromolar range, ~ 100 to $1,000\times$ less sensitive than concentrations of salts for 5-HT secretion from normal and neoplastic EC cells (present study). Irrespective of the reasons for differences in secretion in either intact mucosa or in other enteroendocrine cell lines, the present results indicate the utility of studying isolated normal EC cell and the KRJ-I cell secretory responses *in vitro*. In addition, the differences in response between normal and neoplastic EC cells confirm our hypothesis that neoplastic cells are more sensitive in their response than normal cells to these luminal agents.

To evaluate this difference further, we examined normal and neoplastic EC cell secretory responses to other luminal contents that include trace amines, glucose, bitter tastants, and olfactants, all of which are readily absorbed by the GI tract (37, 49, 83, 87). Trace amines can be produced by bacteria in the gut or are found in some foods, notably chocolate, cheese, and red wine (8). Tyramine is found in micromolar concentrations in cheese and red wine (36), two products associated with paroxysmal symptomatology including sudden diarrhea or meal-ingested flushing in patients with GI carcinoids (56). Both tyramine and octopamine stimulated 5-HT secretion from normal and neoplastic EC cells at levels consistent with concentrations of these agents in the lumen (36), concentrations that correlate with uptake into the GI tract (37, 80). The half-maximal effects in neoplastic cells were measured at 2 – $50\times$ lower concentrations than in normal EC cells, whereas the amplitude of secretion was more than twofold higher in tumor cells. These results indicate that 5-HT secretion from neoplastic EC cells can be induced by foodstuffs associated with the symptoms (secretory diarrhea) of carcinoid disease.

Functional sweet taste receptors responsive to glucose have been identified in duodenal L-cells and provide a novel signaling mechanism that appears capable of regulating intestinal hormone secretion (38). Normal and neoplastic EC cells, whereas expressing transcripts for the sweet TAS1R3 receptor do not express transcripts for its heterodimeric partner, TAS1R2 (51). They do, however, express glucose transporters, GLUT1 and GLUT3, and SGLT1, whose expression is considered to be regulated by TAS1R3 (53). The pancreatic endocrine BON cell line also expresses GLUT1/3 (48). In contrast to this cell line, which responds with 5-HT release to millimolar glucose levels (48), 5-HT release was stimulated in

normal EC cells by both 2-deoxyglucose and sucralose (nanomolar range). This effect could be reversed by phloridzin (half-maximal effect = 0.31–0.35 mM; data not shown). Activation of T1R in L-type cells by sucralose stimulates secretion as well as SGLT1 synthesis leading to glucose absorption and stimulation of GLP-1 and GIP secretion (53). Our results suggest that this occurs in normal EC cells but not in neoplastic cells. Although sucralose stimulated neoplastic 5-HT release, probably through direct activation of T1R (as in L-type cells) (53), 2-deoxyglucose inhibited secretion. This could reflect the very low expression of SGLT1 in neoplastic cells and is supported by observations that 3-methyl-glucose also inhibited neoplastic secretion (data not shown). Nevertheless, an artificial sweetener, sucralose, stimulated 5-HT secretion in both normal and neoplastic EC cells.

Normal and neoplastic cells express bitter G protein-coupled taste receptors. The prototypic bitter tastant, caffeine, acutely (within 30 min of ingestion) increases glucose and insulin secretion in humans, suggesting that luminal sensing of this agent modifies GI hormone secretion (39). In neoplastic EC cells, caffeine stimulates serotonin release with a very low half-maximal effect (subnanomolar) through signal transduction mechanisms that include ERK signaling and calcium flux. Caffeine stimulates both cAMP (3) as well as intracellular calcium release (25), stimulates 5-HT synthesis and levels in the rat brain (73), is a well-known stimulant of intestinal secretion (83), and at high concentrations (30 mM) stimulates 5-HT release into the venous system in perfused rat ilea (71). In normal human EC cells, caffeine stimulates 5-HT release at nanomolar to micromolar levels. Our results confirm that GI neuroendocrine cell 5-HT release can be increased by caffeine, indicating that normal and neoplastic cells are highly sensitive to this agent and provide the basis for understanding the

mechanism by which luminal agents provoke EC cell response and lead to paroxysmal symptomatology.

There are two major classes of olfactant receptors: aquatic (type I) and terrestrial (type II) (72). The class II OR1G1 is the best characterized and is broadly tuned toward odorants of 9–10 carbon chain length, with diverse functional groups (esters, aldehydes, alcohols, and ketones) (72). Little is known regarding the role of this receptor in GI physiology although alterations in calcium flux have been noted in the pancreatic BON cell line (9) in response to some of these agents. Eugenol, a member of the allylbenzene class of chemical compounds, is extracted from cloves and is used as a flavorant, whereas thymol is a monoterpene phenol extracted from thyme and is also used as a flavorant. In the present study, both eugenol and thymol stimulated normal and neoplastic EC cell 5-HT secretion with similar efficacies ($\sim 0.1 \mu\text{M}$). This occurred through activation of calcium influx and was not associated with ERK signaling. The relevance of odorants modifying EC cell secretion emphasizes that these cells are sensors that are capable of detecting a diverse group of luminal ingestants.

Somatostatin receptors are expressed on EC cells, and functional studies have demonstrated that somatostatin analogs inhibit human and rodent EC cell 5-HT secretion (19, 44, 55, 64). In addition, somatostatin analogs are currently the only effective agent to treat neoplastic EC cell secretion (56). Little, however, is known regarding whether somatostatin can affect luminally mediated serotonin release. In the present study, we demonstrate that targeting subtype 2 and 5 somatostatin receptors reverses amino acid, trace amine, tastant (caffeine) and olfactant (thymol)-mediated secretion by a mechanism involving a reduction in ERK signaling. Somatostatin also reduces ERK phosphorylation in pituitary tumors (pituitary adenomas and the rat pituitary cell line GH3), indicating this mechanism

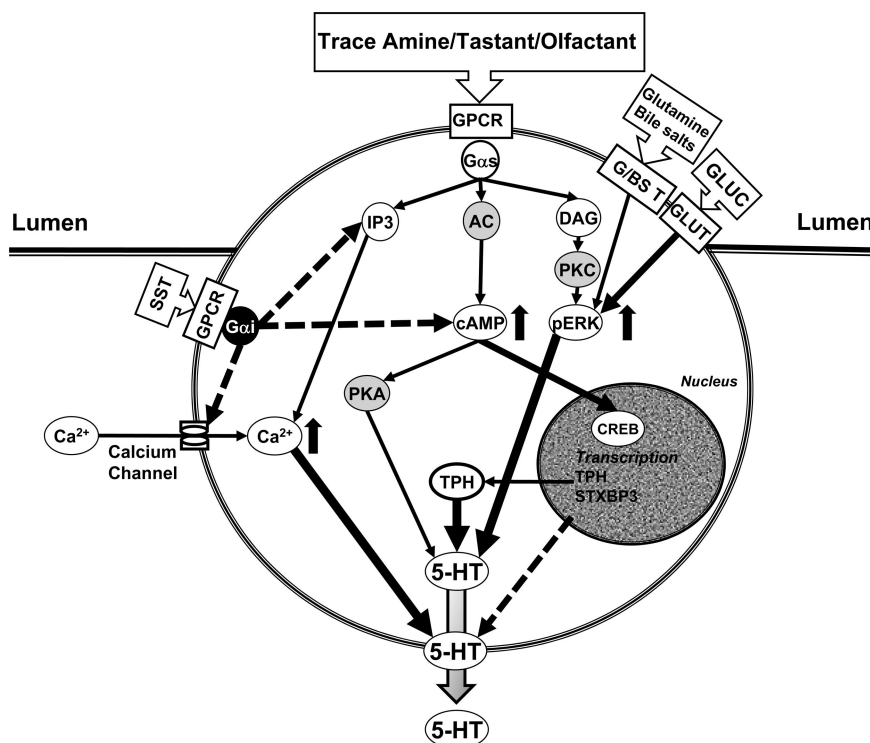


Fig. 10. Model of G protein-coupled receptor (GPCR)-mediated EC cell secretion. Luminally stimulated (e.g., trace amines, tastants, or olfactant) GPCRs induce cAMP/ERK/ Ca^{2+} signaling through activation of adenylyl cyclase (AC), diacylglycerol (DAG), and IP₃ via coupling to G α_s whereas transport of amino acids (e.g., glutamine), bile salts, or glucose into the cell induce ERK phosphorylation. Gene transcription is induced by elevated cAMP through CREB activation of the rate-limiting 5-HT synthesis enzyme tryptophan hydroxylase (TPH) and 11 other genes that represent the secretory transcriptome and includes the vesicle docking inhibitor STXBP3. TPH synthesizes 5-HT, whose secretion is positively regulated through cAMP-activated PKA and pERK (both involved in vesicle maturation and translocation) and through elevated intracellular calcium (Ca^{2+}) (vesicle-membrane docking and exocytosis). Secretion is negatively regulated by somatostatin through G α_i GPCRs, which inhibit calcium influx and ERK phosphorylation. Vesicle docking and exocytosis are directly inhibited by STXBP3, which is therefore also a negative regulator of 5-HT secretion. G/BS T, glutamine/bile salt transporter; GLUC, glucose; GLUT, glucose transporter.

exhibits some degree of commonality in the inhibition of neuroendocrine cell function (35).

Stimulation of neoplastic EC cell secretion is associated with consistent quantitative changes (2- to 3-fold differences) (40, 54) in genes related to protein synthesis and modification (40). Our data demonstrate that glutamine and SDC increase gene expression more than twofold, and we identified 11 protein synthesis/modification as well as trafficking genes involved in secretion. One gene is the syntaxin binding protein 3 (STXBP3). Binding of this protein to Munc18 inhibits the formation of the SNARE complex, which regulates membrane-secretory granule fusion and allows secretion (15, 85). STXBP3 knockdown would theoretically remove this secretory block. In the present studies, targeting STXBP3 increased both basal and β -adrenergic-mediated EC cell secretion, confirming that this gene is involved in the secretory process. This result indicates that targeting the EC cell secretory transcriptome can alter EC cell production and secretion of 5-HT.

In summary, we have identified that luminal amino acids, trace amines, tastants, and olfactants are stimulants of EC cell secretion and demonstrated that neoplastic EC cells are similarly responsive as normal EC cells to these agents except for trace amines and bitter tastants. The mechanisms of this secretion are induced via GPCR-mediated ERK phosphorylation and calcium flux although all tested agents except for caffeine function through only one pathway (Fig. 10). Caffeine activated both signaling pathways and was the most effective EC cell secretagogue. The EC secretory phenotype comprises 11 genes, and targeting expression of STXBP3, which is a critical negative regulator of exocytosis (34), effectively inhibited EC cell secretion. Identifying the luminal factors that regulate EC cell secretion will allow an appreciation of the role of this neuroendocrine cell in abnormal intestinal secretion and motility whereas determining the mechanisms of lumenally mediated secretion will facilitate the development of new agents that selectively target EC cell function.

GRANTS

This work was supported by the Bruggeman Medical Foundation, the Norwegian Medical Research Council, and the Oddrun Mjåland Foundation.

REFERENCES

- Ahlman H. Serotonin and carcinoid tumors. *J Cardiovasc Pharmacol* 7, Suppl 7: S79–S85, 1985.
- Alvarado-Alvarez R, Arechiga H, Garcia U. Serotonin activates a Ca^{2+} -dependent K^{+} current in identified peptidergic neurons from the crayfish. *J Exp Biol* 203: 715–723, 2000.
- Antoni FA, Sosunov AA, Haunso A, Paterson JM, Simpson J. Short-term plasticity of cyclic adenosine 3',5'-monophosphate signaling in anterior pituitary corticotrope cells: the role of adenylyl cyclase isoforms. *Mol Endocrinol* 17: 692–703, 2003.
- Barnett P. Somatostatin and somatostatin receptor physiology. *Endocrine* 20: 255–264, 2003.
- Barter R, Pearse AG. Mammalian enterochromaffin cells as the source of serotonin (5-hydroxytryptamine). *J Pathol Bacteriol* 69: 25–31, 1955.
- Bobker DH, Williams JT. Serotonin-mediated inhibitory postsynaptic potential in guinea-pig prepositus hypoglossi and feedback inhibition by serotonin. *J Physiol* 422: 447–462, 1990.
- Branchek T, Blackburn T. Trace amine receptors as targets for novel therapeutics: legend, myth and fact. *Curr Opin Pharmacol* 3: 90–97, 2003.
- Braun T, Volland P, Kunz L, Prinz C, Gratzl M. Enterochromaffin cells of the human gut: sensors for spices and odorants. *Gastroenterology* 132: 1890–1901, 2007.
- Buchheit KH, Engel G, Mutschler E, Richardson B. Study of the contractile effect of 5-hydroxytryptamine (5-HT) in the isolated longitudinal muscle strip from guinea-pig ileum. Evidence for two distinct release mechanisms. *Naunyn Schmiedeberg's Arch Pharmacol* 329: 36–41, 1985.
- Cetin Y, Kuhn M, Kulaksiz H, Adermann K, Bargsten G, Grube D, Forssmann WG. Enterochromaffin cells of the digestive system: cellular source of guanylin, a guanylate cyclase-activating peptide. *Proc Natl Acad Sci USA* 91: 2935–2939, 1994.
- Challacombe DN, Wheeler EE. Possible neural projections from enterochromaffin cells. *Lancet* 2: 1502, 1983.
- Cooke HJ. Neurotransmitters in neuronal reflexes regulating intestinal secretion. *Ann NY Acad Sci* 915: 77–80, 2000.
- Cooke HJ, Wang YZ, Frieling T, Wood JD. Neural 5-hydroxytryptamine receptors regulate chloride secretion in guinea pig distal colon. *Am J Physiol Gastrointest Liver Physiol* 261: G833–G840, 1991.
- Cosen-Binker LI, Lam PP, Binker MG, Reeve J, Pandol S, Gaisano HY. Alcohol/cholecystokinin-evoked pancreatic acinar basolateral exocytosis is mediated by protein kinase C α phosphorylation of Munc18c. *J Biol Chem* 282: 13047–13058, 2007.
- Coughlin SR, Moskowitz MA, Antoniadis HN, Levine L. Serotonin receptor-mediated stimulation of bovine smooth muscle cell prostacyclin synthesis and its modulation by platelet-derived growth factor. *Proc Natl Acad Sci USA* 78: 7134–7138, 1981.
- Crowell MD. Role of serotonin in the pathophysiology of the irritable bowel syndrome. *Br J Pharmacol* 141: 1285–1293, 2004.
- Evers BM, Townsend CM Jr, Upp JR, Allen E, Hurlbut SC, Kim SW, Rajaraman S, Singh P, Reubi JC, Thompson JC. Establishment and characterization of a human carcinoid in nude mice and effect of various agents on tumor growth. *Gastroenterology* 101: 303–311, 1991.
- Forsberg EJ, Miller RJ. Regulation of serotonin release from rabbit intestinal enterochromaffin cells. *J Pharmacol Exp Ther* 227: 755–766, 1983.
- Freeman S, Bohan D, Darcel N, Raybould H. Luminal glucose sensing in the rat intestine has characteristics of a sodium-glucose co-transporter. *Am J Physiol Gastrointest Liver Physiol* 291: G439–G445, 2006.
- Gavin G, McHenry EW, Wilson MJ. Histamine in canine gastric tissues. *J Physiol* 79: 234–238, 1933.
- Gershon M. Review article: serotonin receptors and transporters—roles in normal and abnormal gastrointestinal motility. *Aliment Pharmacol Ther* 20, Suppl 7: 3–14, 2004.
- Gershon MD. Review article: roles played by 5-hydroxytryptamine in the physiology of the bowel. *Aliment Pharmacol Ther* 13, Suppl 2: 15–30, 1999.
- Gershon MD. Serotonin and its implication for the management of irritable bowel syndrome. *Rev Gastroenterol Disord* 3, Suppl 2: S25–S34, 2003.
- Giraldez T, de la Pena P, Gomez-Varela D, Barros F. Correlation between electrical activity and intracellular Ca^{2+} oscillations in GH3 rat anterior pituitary cells. *Cell Calcium* 31: 65–78, 2002.
- Gustafsson BI, Bakke I, Tommeras K, Waldum HL. A new method for visualization of gut mucosal cells, describing the enterochromaffin cell in the rat gastrointestinal tract. *Scand J Gastroenterol* 41: 390–395, 2006.
- Hagenbuch B, Dawson P. The sodium bile cotransport family SLC10. *Pflügers Arch* 447: 566–570, 2004.
- Hansen MB. The enteric nervous system II: gastrointestinal functions. *Pharmacol Toxicol* 92: 249–257, 2003.
- Heinemann A, Ofner M, Amann R, Peskar BA. A novel assay to measure the calcium flux in human basophils: effects of chemokines and nerve growth factor. *Pharmacology* 67: 49–54, 2003.
- Heitz P, Polak JM, Timson DM, Pearse AG. Enterochromaffin cells as the endocrine source of gastrointestinal substance P. *Histochemistry* 49: 343–347, 1976.
- Higashimoto Y, Opara EC, Liddle RA. Dietary regulation of glucose-dependent insulinotropic peptide (GIP) gene expression in rat small intestine. *Comp Biochem Physiol C Pharmacol Toxicol Endocrinol* 110: 207–214, 1995.
- Hofland LJ, van Hagen PM, Lamberts SW. Functional role of somatostatin receptors in neuroendocrine and immune cells. *Ann Med* 31, Suppl 2: 23–27, 1999.
- Houghton L, Atkinson W, Whitaker R, Whorwell J, Rimmer M. Increased platelet depleted plasma 5-hydroxytryptamine concentrations following meal ingestion in symptomatic female subjects with diarrhea predominant irritable bowel syndrome. *Gut* 52: 663–670, 2003.

34. **Houng A, Polgar J, Reed GL.** Munc18-syntaxin complexes and exocytosis in human platelets. *J Biol Chem* 278: 19627–19633, 2003.
35. **Hubina E, Nanzer AM, Hanson MR, Ciccarelli E, Losa M, Gaia D, Papotti M, Terreni MR, Khalaf S, Jordan S, Czirjak S, Hanzely Z, Nagy GM, Goth MI, Grossman AB, Korbonits M.** Somatostatin analogues stimulate p27 expression and inhibit the MAP kinase pathway in pituitary tumours. *Eur J Endocrinol* 155: 371–379, 2006.
36. **Hyotylainen T, Savola N, Lehtonen P, Riekkola ML.** Determination of biogenic amines in wine by multidimensional liquid chromatography with online derivatisation. *Analyst* 126: 2124–2127, 2001.
37. **Iseki K, Sugawara M, Saitoh N, Miyazaki K.** The transport mechanisms of organic cations and their zwitterionic derivatives across rat intestinal brush-border membrane. II. Comparison of the membrane potential effect on the uptake by membrane vesicles. *Biochim Biophys Acta* 1152: 9–14, 1993.
38. **Jang HJ, Kokrashvili Z, Theodorakis MJ, Carlson OD, Kim BJ, Zhou J, Kim HH, Xu X, Chan SL, Juhaszova M, Bernier M, Mosinger B, Margolske RF, Egan JM.** Gut-expressed gustducin and taste receptors regulate secretion of glucagon-like peptide-1. *Proc Natl Acad Sci USA* 104: 15069–15074, 2007.
39. **Johnston KL, Clifford MN, Morgan LM.** Coffee acutely modifies gastrointestinal hormone secretion and glucose tolerance in humans: glycemic effects of chlorogenic acid and caffeine. *Am J Clin Nutr* 78: 728–733, 2003.
40. **Jura J, Wegrzyn P, Zarebski A, Wladyka B, Koj A.** Identification of changes in the transcriptome profile of human hepatoma He pg2 cells stimulated with interleukin-1 beta. *Biochim Biophys Acta* 1689: 120–133, 2004.
41. **Katsuma S, Hirasawa A, Tsujimoto G.** Bile acids promote glucagon-like peptide-1 secretion through TGR5 in a murine enteroendocrine cell line STC-1. *Biochem Biophys Res Commun* 329: 386–390, 2005.
42. **Kellum JM, Donowitz M, Cerel A, Wu J.** Acid and isoproterenol cause serotonin release by acting on opposite surfaces of duodenal mucosa. *J Surg Res* 36: 172–176, 1984.
43. **Kidd M, Eick GN, Modlin IM, Pfragner R, Champaneria MC, Murren J.** Further delineation of the continuous human neoplastic enterochromaffin cell line, KRJ-1, and the inhibitory effects of lanreotide and rapamycin. *J Mol Endocrinol* 38: 181–192, 2007.
44. **Kidd M, Modlin I, Eick G, Champaneria M.** Isolation, purification and functional characterization of the Mastomys EC cell. *Am J Physiol Gastrointest Liver Physiol* 291: G778–G791, 2006.
45. **Kidd M, Modlin IM, Eick GN, Camp RL, Mane SM.** Role of CCN2/CTGF in the proliferation of *Mastomys* enterochromaffin-like cells and gastric carcinoid development. *Am J Physiol Gastrointest Liver Physiol* 292: G191–G200, 2007.
46. **Kidd M, Nadler B, Mane SM, Eick GN, Malfetheriner MV, Champaneria MC, Pfragner R, Modlin IM.** GeneChip, geNorm and gastrointestinal tumors: novel reference genes for real-time PCR. *Physiol Genomics* 30: 363–370, 2007.
47. **Kidd M, Schally AV, Pfragner R, Malfetheriner MV, Modlin IM.** Inhibition of proliferation of small intestinal and bronchopulmonary neuroendocrine cell lines by using peptide analogs targeting receptors. *Cancer* 112: 1404–1414, 2008.
48. **Kim M, Cooke H, Javed N, Carey H, Christofi F, Raybould H.** D-Glucose releases 5-hydroxytryptamine from human BON cells as a model of enterochromaffin cells. *Gastroenterology* 121: 1400–1406, 2001.
49. **Kuwayama K, Inoue H, Kanamori T, Tsujikawa K, Miyaguchi H, Iwata Y, Miyauchi S, Kamo N, Kishi T.** Interactions between 3,4-methylenedioxymethamphetamine, methamphetamine, ketamine, and caffeine in human intestinal Caco-2 cells and in oral administration to rats. *Forensic Sci Int* 170: 183–188, 2007.
50. **Lee YC, Asa SL, Drucker DJ.** Glucagon gene 5'-flanking sequences direct expression of simian virus 40 large T antigen to the intestine, producing carcinoma of the large bowel in transgenic mice. *J Biol Chem* 267: 10705–10708, 1992.
51. **Li X, Staszewski L, Xu H, Durick K, Zoller M, Adler E.** Human receptors for sweet and umami taste. *Proc Natl Acad Sci USA* 99: 4692–4696, 2002.
52. **Mackenzie B, Erickson J.** Sodium-coupled neutral amino acid (System N/A) transporters of the SLC38 gene family. *Pflügers Arch* 447: 784–795, 2004.
53. **Margolske RF, Dyer J, Kokrashvili Z, Salmon KS, Ilegems E, Daly K, Maillet EL, Ninomiya Y, Mosinger B, Shirazi-Beechey SP.** T1R3 and gustducin in gut sense sugars to regulate expression of Na⁺-glucose cotransporter 1. *Proc Natl Acad Sci USA* 104: 15075–15080, 2007.
54. **Marti M, Regos A, Li Y, Schraner E, Wild P, Muller N, Knopf L, Hehl A.** An ancestral secretory in the protozoan parasite *Giardia intestinalis*. *J Biol Chem* 278: 24837–24848, 2003.
55. **Modlin IM, Kidd M, Eick G, Champaneria M.** The functional characterization of normal and neoplastic EC cells. *J Clin Endocrinol Metab* 91: 2340–2348, 2006.
56. **Modlin IM, Kidd M, Latich I, Zikusoka MN, Shapiro MD.** Current status of gastrointestinal carcinoids. *Gastroenterology* 128: 1717–1751, 2005.
57. **Modlin IM, Latich I, Kidd M, Zikusoka M, Eick G.** Therapeutic options for gastrointestinal carcinoids. *Clin Gastroenterol Hepatol* 4: 526–547, 2006.
58. **Nilsson O, Ahlman H, Geffard M, Dahlstrom A, Ericson LE.** Bipolarity of duodenal enterochromaffin cells in the rat. *Cell Tissue Res* 248: 49–54, 1987.
59. **Nilsson O, Ericson LE, Dahlstrom A, Ekholm R, Steinbusch HW, Ahlman H.** Subcellular localization of serotonin immunoreactivity in rat enterochromaffin cells. *Histochemistry* 82: 351–355, 1985.
- 59a. **Partek Genomics Suite.** St. Louis, MO: Partek Inc., 2008.
60. **Pearse AG, Polak JM, Bloom SR, Adams C, Dryburgh JR, Brown JC.** Enterochromaffin cells of the mammalian small intestine as the source of motilin. *Virchows Arch* 16: 111–120, 1974.
61. **Peregrin A, Ahlman H, Jodal M, Lundgren O.** Involvement of serotonin and calcium channels in the intestinal fluid secretion evoked by bile salt and cholera toxin. *Br J Pharmacol* 127: 887–894, 1999.
62. **Pfragner R, Wirnsberger G, Niederle B, Behmel A, Rinner I, Mandl A, Wawrina F, Luo J, Adamiker D, Hoyer H, Ingolic E, Schauenstein K.** Establishment of a continuous cell line from a human carcinoid of the small intestine (KRJ-1): characterization of the effects of 5-azacytidine on proliferation. *Int J Oncol* 8: 513–520, 1996.
63. **Podolsky DK.** Inflammatory bowel disease. *N Engl J Med* 347: 417–429, 2002.
64. **Racke K, Reimann A, Schworer H, Kilbinger H.** Regulation of 5-HT release from enterochromaffin cells. *Behav Brain Res* 73: 83–87, 1996.
65. **Racke K, Schworer H.** Nicotinic and muscarinic modulation of 5-hydroxytryptamine (5-HT) release from porcine and canine small intestine. *Clin Invest* 70: 190–200, 1992.
66. **Racke K, Schworer H, Kilbinger H.** Adrenergic modulation of the release of 5-hydroxytryptamine from the vascularly perfused ileum of the guinea-pig. *Br J Pharmacol* 95: 923–931, 1988.
67. **Raybould H, Cooke H, Christof F.** Sensory mechanisms: transmitters, modulators and reflexes. *Neurogastroenterol Motil* 16, Suppl 1: 60–63, 2004.
68. **Reimann F, Williams L, da Silva Xavier G., Rutter GA, Gribble FM.** Glutamine potently stimulates glucagon-like peptide-1 secretion from GLUTag cells. *Diabetologia* 47: 1592–1601, 2004.
69. **Reimer RA.** Meat hydrolysate and essential amino acid-induced glucagon-like peptide-1 secretion, in the human NCI-H716 enteroendocrine cell line, is regulated by extracellular signal-regulated kinase1/2 and p38 mitogen-activated protein kinases. *J Endocrinol* 191: 159–170, 2006.
70. **Reynolds A, Anderson EM, Vermeulen A, Fedorov Y, Robinson K, Leake D, Karpilow J, Marshall WS, Khvorova A.** Induction of the interferon response by siRNA is cell type- and duplex length-dependent. *RNA* 12: 988–993, 2006.
71. **Rezaie-Majd S, Murar J, Nelson DP, Kelly RF, Hong Z, Lang IM, Varghese A, Weir EK.** Increased release of serotonin from rat ileum due to dexfenfluramine. *Am J Physiol Regul Integr Comp Physiol* 287: R1209–R1213, 2004.
72. **Sanz G, Schlegel C, Pernollet JC, Briand L.** Comparison of odorant specificity of two human olfactory receptors from different phylogenetic classes and evidence for antagonism. *Chem Senses* 30: 69–80, 2005.
73. **Schlossberg AJ.** Acute and chronic effects of caffeine on brain monoamine levels and endocrine function in the rat. *Arch Int Pharmacodyn Ther* 267: 149–160, 1984.
74. **Sidhu M, Cooke HJ.** Role for 5-HT and ACh in submucosal reflexes mediating colonic secretion. *Am J Physiol Gastrointest Liver Physiol* 269: G346–G351, 1995.
75. **Siriwardena A, Kellum JM Jr.** A 5-HT₂ receptor mediates serotonin-induced electrolyte transport in rat left colon. *J Surg Res* 55: 323–329, 1993.

76. Soeters PB, van de Poll MC, van Gemert WG, Dejong CH. Amino acid adequacy in pathophysiological states. *J Nutr* 134: 1575S–1582S, 2004.
77. Spiller R. Role of motility in chronic diarrhea. *Neurogastroenterol Motil* 18: 1045–1055, 2006.
78. Sternini C. Taste receptors in the gastrointestinal tract. IV. Functional implications of bitter taste receptors in gastrointestinal chemosensing. *Am J Physiol Gastrointest Liver Physiol* 292: G457–G461, 2007.
79. Sutherland K, Young RL, Cooper NJ, Horowitz M, Blackshaw LA. Phenotypic characterization of taste cells of the mouse small intestine. *Am J Physiol Gastrointest Liver Physiol* 292: G1420–G1428, 2007.
80. Tcheransky DM, Acevedo C, Rubio MC. Studies of tyramine transfer and metabolism using an in vitro intestinal preparation. *J Pharm Sci* 83: 549–552, 1994.
81. Van der Horst-Schrivers AN, Wymenga AN, Links TP, Willemse PH, Kema IP, de Vries EG. Complications of midgut carcinoid tumors and carcinoid syndrome. *Neuroendocrinology* 80, Suppl 1: 28–32, 2004.
82. Vandesompele J, De Preter K, Pattyn F, Poppe B, Van Roy N, De Paepe A, Speleman F. Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biol* 3: RESEARCH0034, 2002.
83. Wald A, Back C, Bayless TM. Effect of caffeine on the human small intestine. *Gastroenterology* 71: 738–742, 1976.
84. Wang Y, Prpic V, Green GM, Reeve JR Jr, Liddle RA. Luminal CCK-releasing factor stimulates CCK release from human intestinal endocrine and STC-1 cells. *Am J Physiol Gastrointest Liver Physiol* 282: G16–G22, 2002.
85. Wheeler MB, Sheu L, Ghai M, Bouquillon A, Grondin G, Weller U, Beaudoin AR, Bennett MK, Trimble WS, Gaisano HY. Characterization of SNARE protein expression in beta cell lines and pancreatic islets. *Endocrinology* 137: 1340–1348, 1996.
86. Wischmeyer P. Glutamine: role in gut protection in critical illnesses. *Curr Opin Clin Nutr Metab Care* 9: 607–612, 2006.
87. Zaloga GP, Siddiqui RA. Biologically active dietary peptides. *Mini Rev Med Chem* 4: 815–821, 2004.
88. Zhu J, Wu X, Owyang C, Li Y. Intestinal serotonin acts as a paracrine substance to mediate vagal signal transmission evoked by luminal factors in the rat. *J Physiol* 530: 431–442, 2001.

